

Fence Panel Load Capacity Data

Structural performance of KESTREL 3D welded panels & post systems

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1. Panel Geometry Tested

Model	Size (mm)	Wire (mm)	Mesh (mm)	V-beams
KM-3D-24	2400 × 2030	5 / 6	50 × 200	4
KM-3D-25	2500 × 2500	5 / 6	50 × 200	5
KM-3D-33	3300 × 2030	6 / 7	50 × 200	7

2. Wind Load Performance (panel + post, post spacing 2.5 m)

Panel Height	Post Section	Max Wind Load	Deflection @ Max
1.03 m	60×40×1.5 mm	2.4 kN/m²	< L/150
1.73 m	60×40×1.5 mm	1.7 kN/m²	< L/150
2.43 m	80×60×2.0 mm	1.2 kN/m²	< L/150

3. Static Point Load (mid-span, 2.03 m span)

Wire Config.	Permanent Set Limit	Measured Load	Safety Factor
4 mm / 4 mm	0.5 kN	0.62 kN	1.24
5 mm / 6 mm	0.5 kN	0.86 kN	1.72
6 mm / 7 mm	0.5 kN	1.05 kN	2.10

4. Test Methods

- Wind load: uniform pressure rig per EN 13241-1 §4.3.2, increments of 0.1 kN/m²
- Point load: EN 13241-3 soft body impact + rigid body static test
- Panel strength: line load at 100 mm above bottom rail, rate 10 mm/min

Site-specific design (post spacing, foundation, wind zone) must be verified by the responsible engineer. Free calculation assistance available from our technical desk.